`dr-iissi` Theme Guide

Usage reference and visual regression deck

IISSI Marp theme

UNIVERSIDAD D SEVILLA

What this deck is for

Learn how to use the `dr-iissi` theme in real slides.

See the supported layout patterns and content conventions.

Verify typography, code blocks, math, image framing, footer, and pagination.

Use the later sections as a visual regression deck after theme or engine changes.

Quick start

Use the theme with Marp CLI:

```
marp --theme-set ./css/dr-iissi.css --engine ./scripts/marp-engine.js slide.md --pdf
```

Recommended front matter in a deck that lives outside `marp-theme/`:

```
---
marp: true
theme: dr-iissi
marp-theme-rel-dir: ../../marp-theme
paginate: true
size: 16:9
image-box: false
---
```

Theme-specific front matter

`marp-theme-rel-dir` tells the engine where theme assets live relative to the current markdown file.

- In this guide the value is `..`
- In `iissi-1/t*/...md` files the value is `../../marp-theme`
- Use a relative path so PDF builds stay portable across machines

`image-box` controls whether images use the default framed box style.

- Use `image-box: false` to disable the image box for the whole deck
- Omit it, or use `image-box: true`, to keep the default behavior

By default, images use a very subtle granate frame.

If the path is wrong, background images and local fonts may not resolve during render.

Column marker rules

For `two-col`, `two-col-3-7`, `two-col-7-3`, and `three-col`:

1. Use `*** left ***` to start the left column.
2. Use `*** right ***` to start the right column.
3. Use `*** center ***` only in `three-col`.
4. Do not use `<hr/>` inside column layouts.

Optional: add `v-align` to vertically center the slide body.

The engine inserts the needed blank line after each column marker automatically, so the next block can start immediately.

Guide map

Section 1. Typography and code fonts

Section 2. Base layouts and images

Section 3. Vertical alignment

Section 4. `mr`, `sql`, and KaTeX

Section 5. Visual regression cases A-J

Section 6. Realistic topic examples T1-T12

Section 7. Mixed regression cases K-N

Theme font: default (`Roboto`)

This slide uses the default theme font.

- Accents and symbols: áéíóú ñ ç
- **Bold sample**
- *Italic sample*
- `Inline code sample`

Theme font override: `Lato`

Use slide classes such as `font-lato` when a deck needs a different tone without changing the whole theme.

- Paragraph rhythm sample
- **Bold sample**
- *Italic sample*
- `Inline code sample`

Theme font override: `Roboto Condensed`

This variant is useful for denser titles or compact text blocks.

- Narrower letterforms
- Good for labels and summaries
- `Inline code sample`

Theme font override: `Source Sans 3`

Use this slide to validate the local variable font and italic face.

- Normal text sample
- *Italic sample*
- **Bold sample**

Theme font override: `Caveat`

This slide is intentionally distinctive so font loading issues are obvious.

- Handwritten tone sample
- Long enough sentence to inspect rhythm and contrast.
- **Bold sample**
- *Italic sample*

Code font: default stack

Default code blocks use the theme monospace stack.

```
SELECT
  d.nombre AS departamento,
  e.id_emp,
  e.nombre,
  e.salario,
  ROW_NUMBER() OVER (
    PARTITION BY d.id_dept
    ORDER BY e.salario DESC
  ) AS orden_salario
FROM Empleado e
JOIN Departamento d ON d.id_dept = e.id_dept
WHERE e.salario IS NOT NULL
ORDER BY d.nombre, orden_salario;
```

Code font: `JetBrains Mono`

The engine supports a local JetBrains Mono variant through the fence info string.

```
Empleado(PK id_emp, nombre, salario, FK id_dept null)
Departamento(PK id_dept, nombre AK, presupuesto)
Proyecto(PK id_proy, nombre, FK id_dept)
Asignacion(PK id_emp, PK id_proy, horas, fecha_inicio)
```

```sql
SELECT e.nombre, p.nombre, a.horas
FROM Asignacion a
JOIN Empleado e ON e.id_emp = a.id_emp
JOIN Proyecto p ON p.id_proy = a.id_proy
WHERE a.horas ≥ 20
ORDER BY a.horas DESC, e.nombre;
```

Code fence selectors

Pick the code font in the fence info string:

- `default` or `consolas` for the default monospace stack
- `jetbrains` or `jb` for `JetBrains Mono`

Examples:

```
```sql default
SELECT * FROM Empleado;
```
```

```
```mr jetbrains
Empleado(PK id_emp, nombre)
```
```

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Layout: `default`

Standard single-column content slide.

- Main point one
- Main point two
- Subpoints can be nested
- Suitable for most explanatory slides

Headings: `h2`, `h3`, `h4`

This is an `h2`

This is an `h3`

This is an `h4`

Body text should remain readable below centered secondary headings.

Layout: `content`

> Use this style for agendas and section summaries

> The layout is intentionally lighter than `default`

> It works well with short, scannable bullet lists

Layout: `default` with a full image

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Layout: `default` with text and image

- Text may appear before a framed image.
- Images can keep their box styling, or disable it with `image-box: false`.
- Spacing should remain consistent.

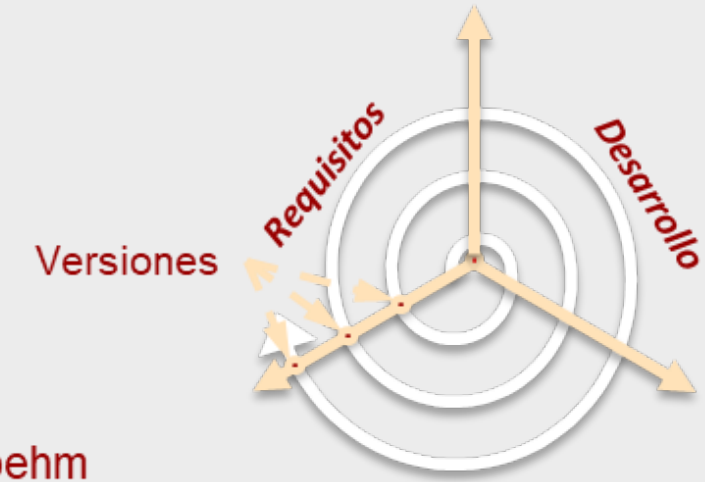

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Image box option

Use `image-box: false` in the YAML front matter to remove the default framed image box for the whole deck.

Example:

```
---
marp: true
theme: dr-iissi
marp-theme-rel-dir: ..
paginate: true
image-box: false
---
```

This deck normally renders with the default image box enabled, so visual checks for `image-box: false` should be done by temporarily enabling that option in this file.

Layout: `two-col`

Left column

Content for the first half.

- Item A
- Item B
- Item C

Right column

Content for the second half.

- Item D
- Item E
- Item F

Layout: `three-col`

First column

Text for column one.

Second column

Text for column two.

Third column

Text for column three.

Layout: text + code + image

Quick summary:

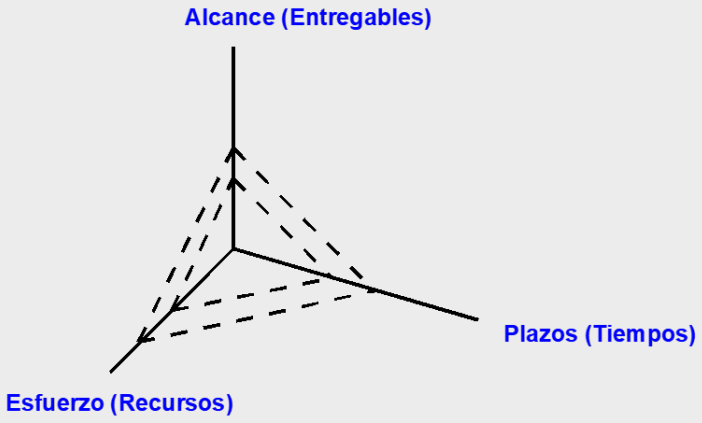
- Context
- Rule
- Conclusion

```
SELECT nombre, salario
FROM Empleados
WHERE salario > 2000;
```

Layout: `three-col v-align`

Vertically centered text.

```
Pedido(PK id_ped, FK id_cli)
Cliente(PK id_cli, nombre)
```



Layout: `two-col-3-7`

Left column at 30%.

Short contextual content.

The right column gets the main space.

- Detailed point one
- Detailed point two
- Detailed point three

Layout: `two-col-7-3`

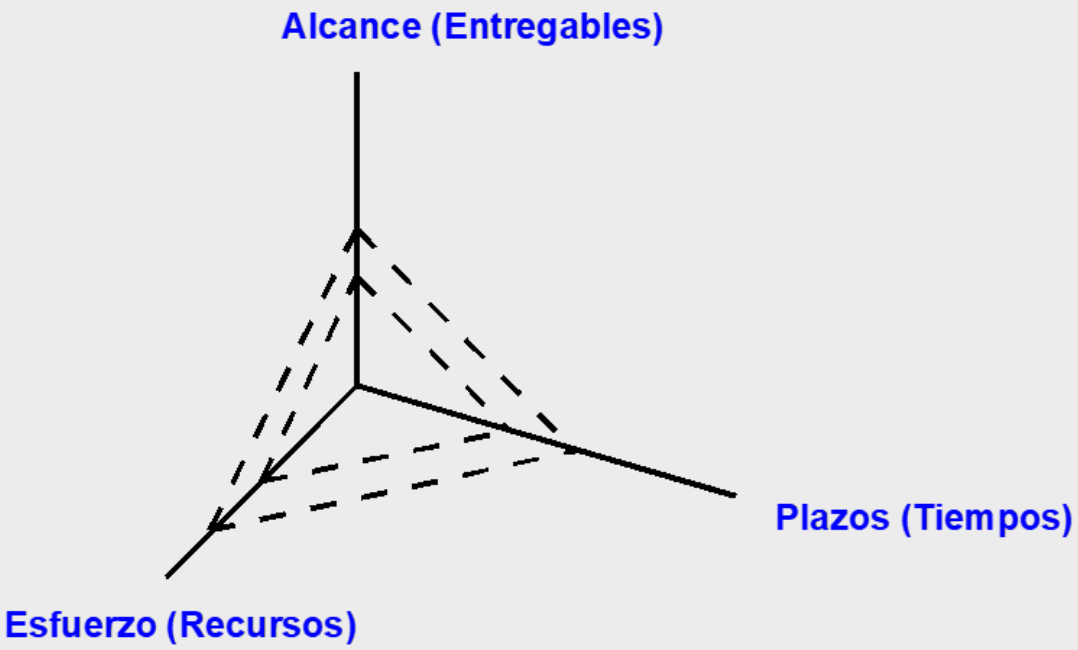
The left column holds the main explanation.

- Main concept
- Supporting point
- Partial conclusion

The right column is suitable for notes or compact media.

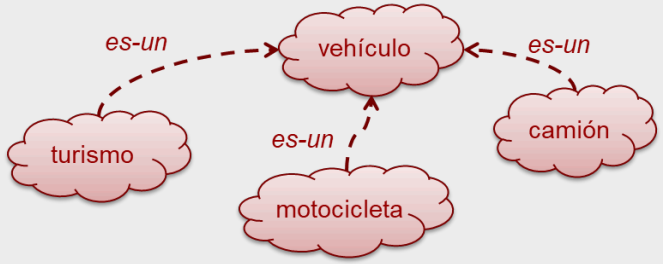
Layout: image and text in 50/50

Text above an image.



- Feature A
- Feature B

Layout: image-heavy `two-col-3-7`



Caption-like text
below the image.

- More detailed explanation
- Supporting note
- Closing point

Layout: text opposite image

Text at the left side of the image.

- Feature A
- Feature B

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Layout: text-heavy `two-col-7-3`

Primary explanation with more horizontal room.

- Point one
- Point two
- Point three

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`v-align` on `default`

This block should appear vertically centered inside the slide body.

`v-align` on `content`

Item 1

Item 2

Item 3

`v-align` on `two-col`

Text in the left column.

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`v-align` on `two-col-3-7`

Short text.

```
SELECT *
FROM Empleados
WHERE salario > 2000;
```

`v-align` on `two-col-7-3`

```
Empleado(PK id_emp, nombre, FK id_dept)
Departamento(PK id_dept, nombre)
```

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`mr` code blocks

Minimal relational-model sample used to validate custom highlighting.

```
Empleado(PK id_emp, nombre, apellidos, salario, FK id_dept null)
Departamento(PK id_dept, nombre AK, presupuesto)
Proyecto(PK id_proy, nombre, FK id_dept)
Asignacion(PK id_emp, PK id_proy, horas, fecha_inicio)
```

`sql` code blocks

Longer SQL sample used to inspect indentation, wrapping, and readability.

```sql
SELECT
  e.id_emp,
  e.nombre,
  d.nombre AS departamento,
  e.salario
FROM Empleado e
JOIN Departamento d ON e.id_dept = d.id_dept
WHERE e.salario > 30000
  AND d.presupuesto >= 100000
ORDER BY d.nombre, e.salario DESC, e.nombre;
```

`python`, `javascript`, `html`, and `css` code blocks

Same code-block treatment as `mr` and `sql`, with distinct background colors.

```python
def salario_neto(bruto, retencion=0.15):
  return round(bruto * (1 - retencion), 2)
```

```javascript
const empleados = datos.filter((e) => e.salario > 30000);
console.log(empleados.map((e) => e.nombre));
```

```html
<section class="empleados">
  <h2>Departamento</h2>
  <p>Listado actualizado</p>
</section>
```

```css
.empleados {
  border: 2px solid #9f1732;
  padding: 0.75rem 1rem;
  background: #fff;
}
```

`mr` + `sql` in columns

```
Empleado(PK id_emp, nombre, salario,
Departamento(PK id_dept, nombre AK, p
Asignacion(PK id_emp, PK id_proy, hor
```

```sql
SELECT
  e.nombre,
  d.nombre AS departamento,
  SUM(a.horas) AS total_horas
FROM Empleado e
JOIN Departamento d ON d.id_dept = e.id_dept
LEFT JOIN Asignacion a ON a.id_emp = e.id_emp
WHERE e.id_dept IS NOT NULL
GROUP BY e.nombre, d.nombre
HAVING SUM(a.horas) > 10
ORDER BY total_horas DESC, e.nombre;
```

KaTeX formulas

$$\prod_{\text{columns}} \left(\sigma_{\text{condition}} (T_1 \times T_2 \times \cdots \times T_n) \right)$$

```
SELECT <column list>
FROM <T1, T2, ..., Tn>
WHERE <condition>
```

$$\prod_{\text{name,salary}} \left(\sigma_{\text{salary}<2000} (Empleados) \right)$$

```
SELECT nombre, salario
FROM Empleados
WHERE salario < 2000;
```

$$\underset{\text{P.de}}{\Upsilon}^{\text{P.de, count(Ped.id)}} (Ped \bowtie P)$$

Algebra macros reference

Write display formulas with `$$ ... $$` and use:

- `\Proj_{...}` for projection
- `\Sel_{...}` for selection
- `\Ren_{...}` for renaming
- `\Group^{...}_{...}` for grouping and aggregation
- `\NatJoin`, `\JoinBy{...}` for joins
- `\Union`, `\Inter`, `\Diff` for set operators

Algebra macros source

```
\Proj_{nombre,salario}(Empleados)
\Sel_{salario>2000}(Empleados)
\Ren_{Emp}(Empleados)
\Group^{count(Ped.id)}_{P.de}(Ped \NatJoin P)
\Sel_{Emp.id=Dept.id}(Emp \JoinBy{\text{id}} Dept)
```

Algebra macros rendered

$$\prod_{\text{nombre,salario}} (Empleados)$$

$$\sigma_{\text{salario}>2000} (Empleados)$$

$$\rho_{\text{Emp}} (Empleados)$$

$$\Upsilon^{\text{count}(Ped.id)}_{\text{P.de}} (Ped \bowtie P)$$

$$Emp \cup Dept, \quad Emp \cap Dept, \quad Emp \setminus Dept$$

Special characters and icons

Arrow examples: ➡️ ⬅️

Icon examples: 🎯🍽️🍴🧾

Algebra symbols: π σ ρ ≥ γ ∧ ∨

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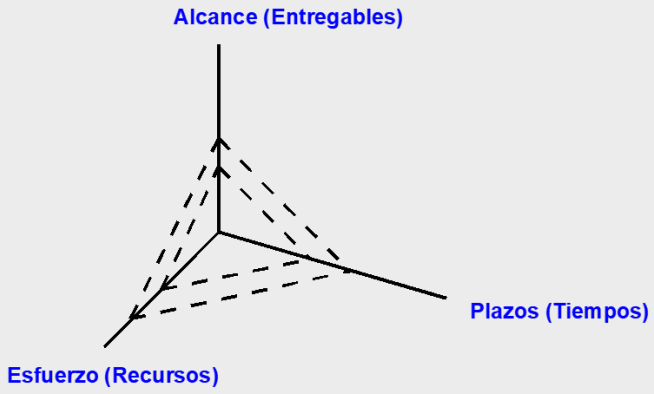
Visual regression coverage A-J

These slides are intended as regression probes.

They cover:

- image in left, right, or both columns
- text above and below framed images
- `two-col`, `two-col-3-7`, and `two-col-7-3`
- mixed cases with code and images
- image sizing and clipping behavior inside columns

Test A: left image only

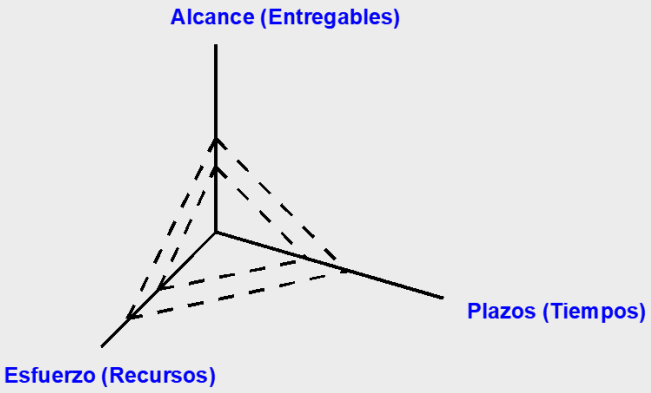


Regular text in the right column to force height distribution.

- Line 1
- Line 2
- Line 3

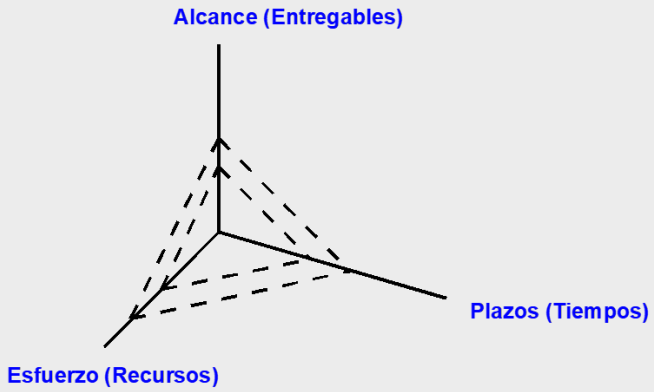
Test B: left image with text above

Text above the image.



Control content in the right column.

Test C: left image with text below

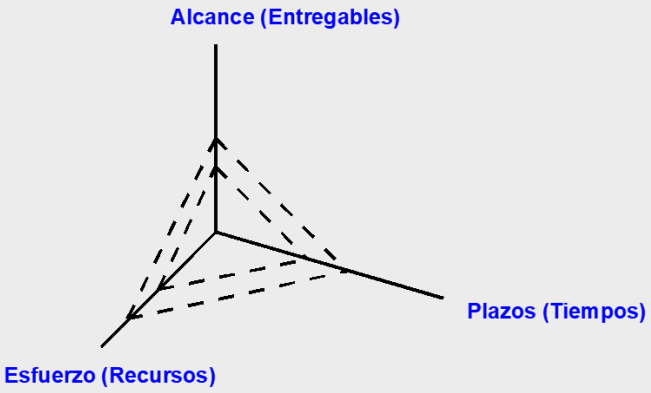


Text below the image.

Control content in the right column.

Test D: left image with text above and below

Text above the image.



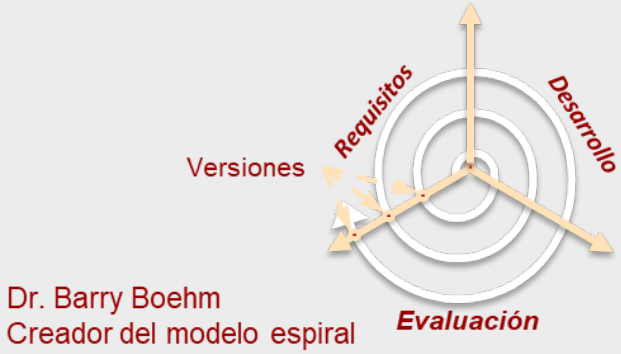
Text below the image.

Control content in the right column.

Test E: right image only

Regular text in the left column to force height distribution.

- Line 1
- Line 2
- Line 3
- Line 4



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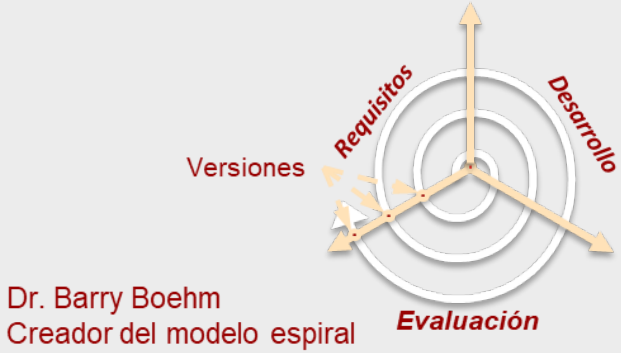
Test F: right image with text above and below

Text on the left.

Text above the image.



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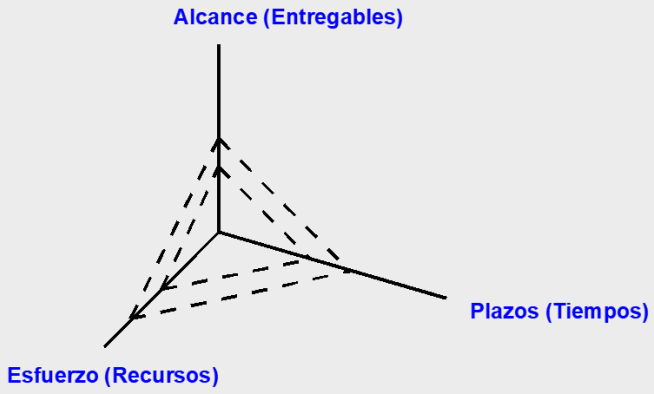
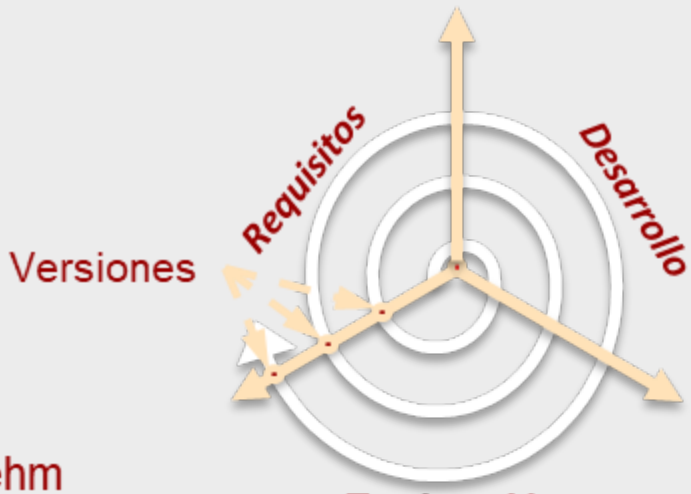
Text below the image.

Test G: image in both columns



Text below the left image.

Text above the right image.



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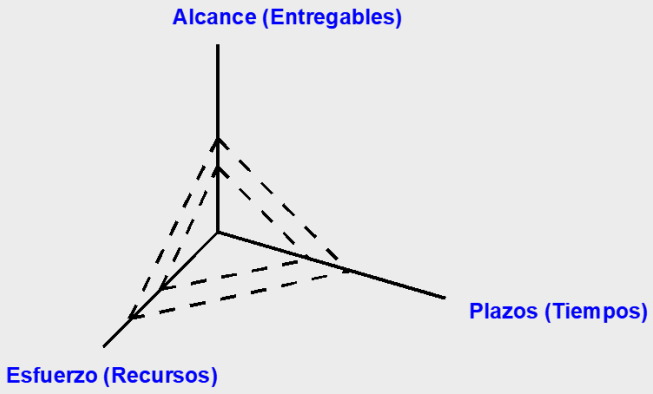
Test H: image in both columns, inverted

Text above the left image.



Text below the right image.

Test I: 50/50 with two images and text

Text above the left image.

Text below the left image.

Text above the right image.

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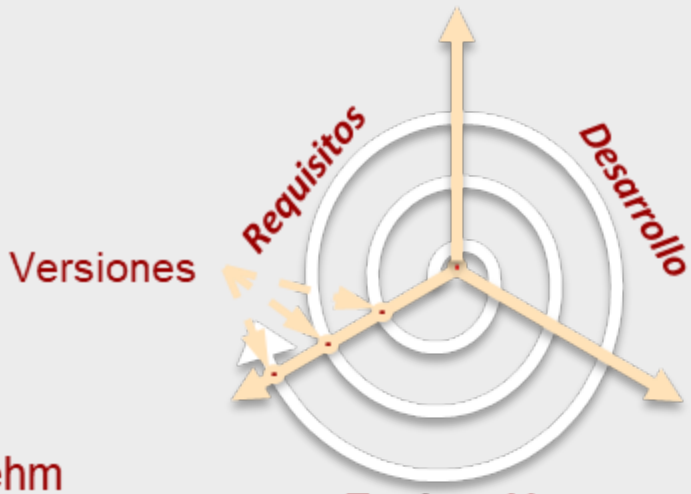
Text below the right image.

Test J: 50/50 with two images and text, inverted

Text above the left image.



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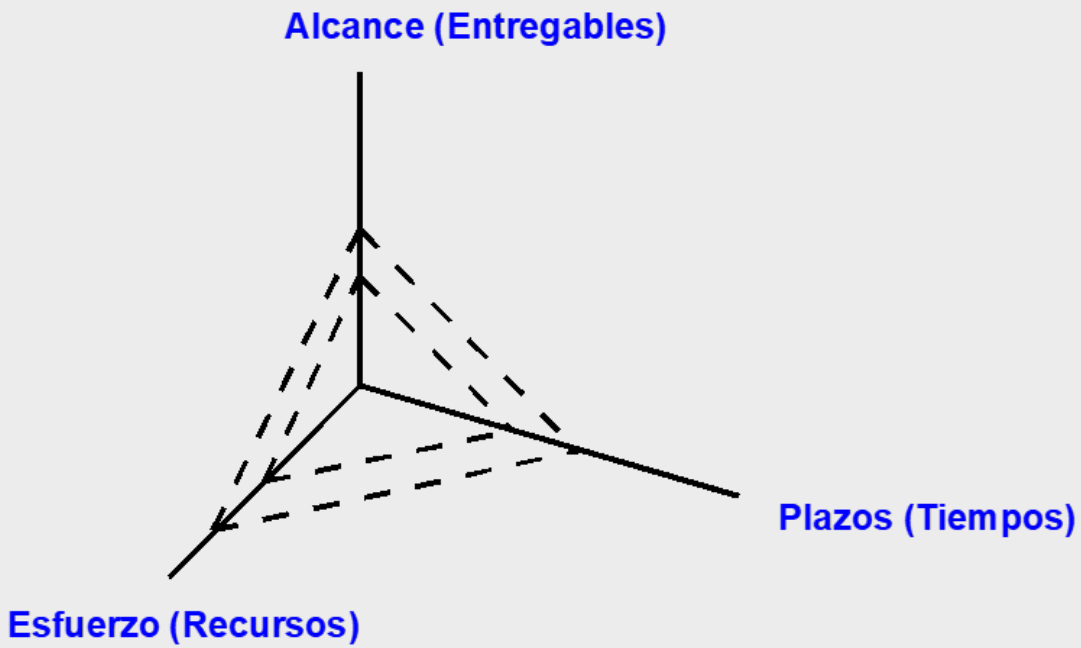
Text below the left image.

Text above the right image.



Text below the right image.

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Realistic topic examples T1-T12

These slides are not only usage examples.

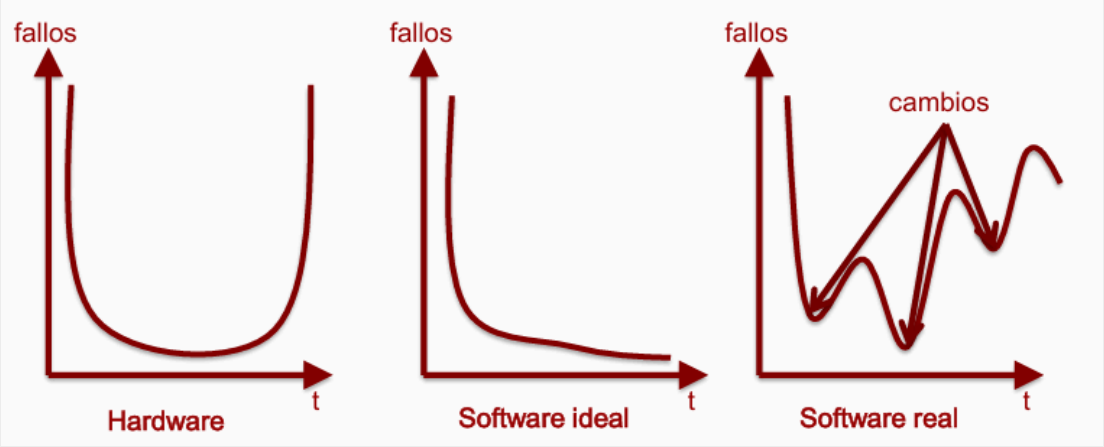
They also validate that the theme remains usable with realistic content taken from the course topics:

- image-driven slides
- asymmetric columns
- `mr` and `sql` blocks
- formulas and diagrams

T1: software fundamentals

The software crisis appears when:

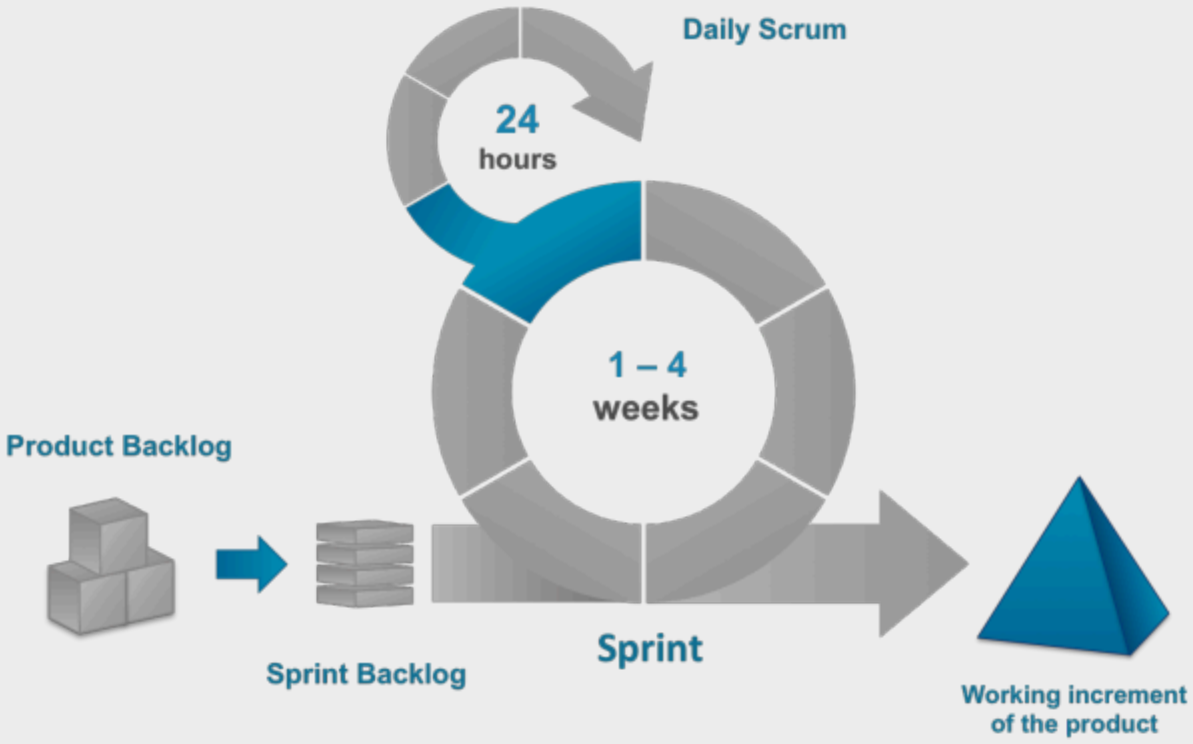
- projects become larger,
- technical complexity grows,
- engineering processes remain unclear.



T2: Scrum overview

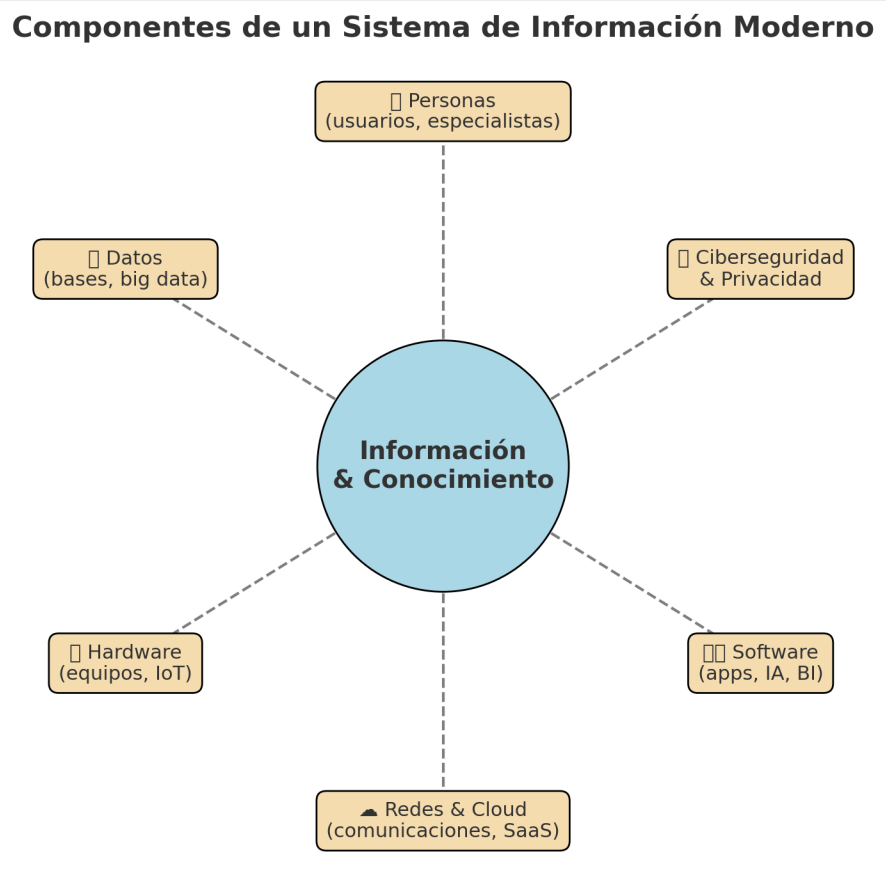
Main roles and artifacts:

- `Product Owner`
- `Scrum Master`
- `Sprint Backlog`



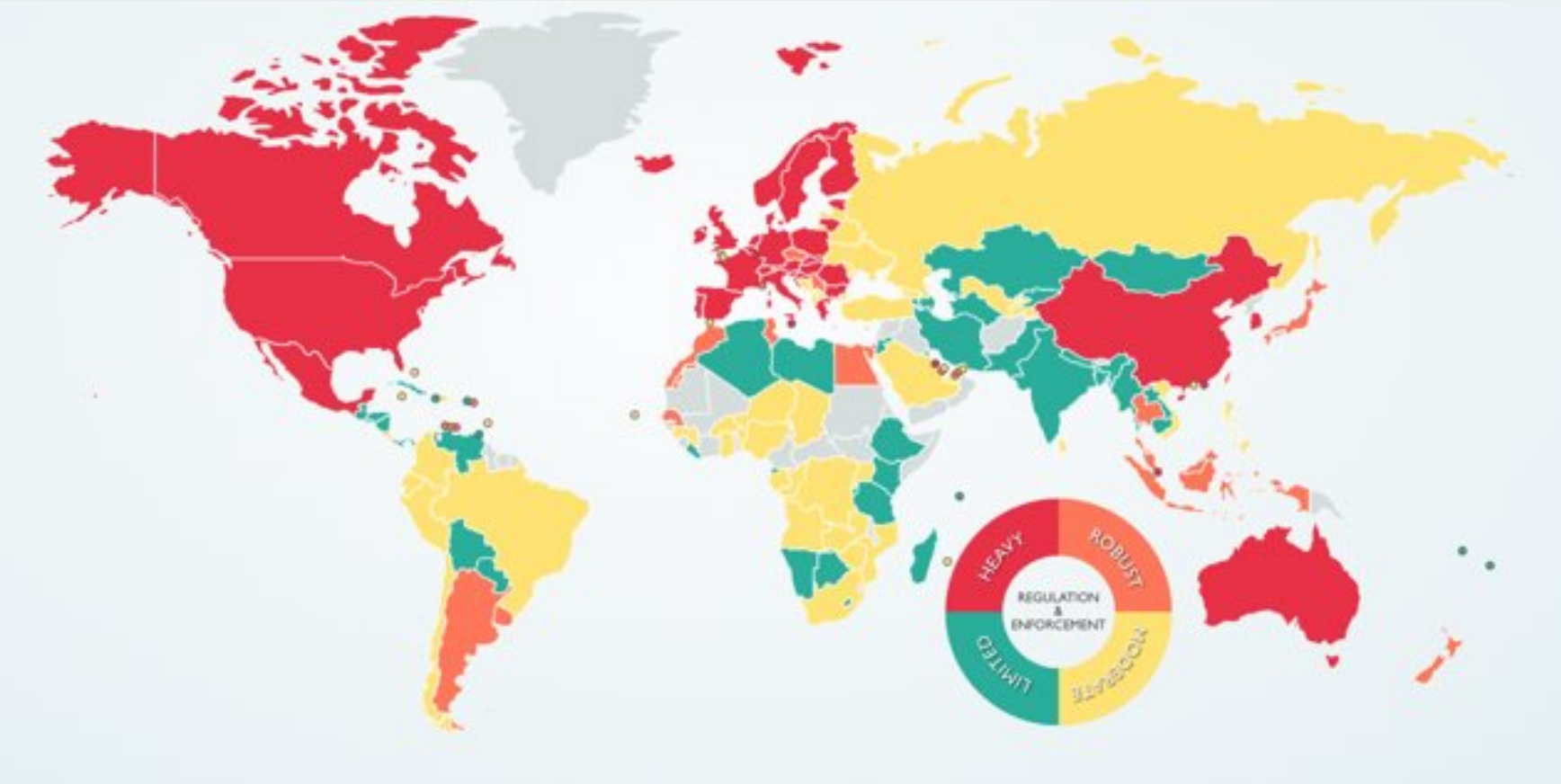
T3: information system components

An information system combines people, processes, data, and technology.



T4: data protection

Global overview of data protection regulation.



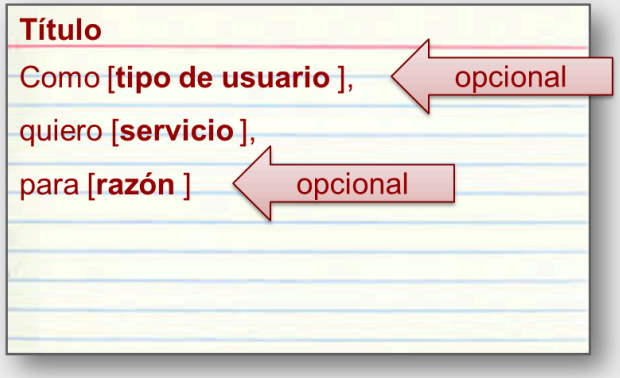
T5: user stories

Base structure:

> As a `<role>`, I want `<goal>` so that `<benefit>`.

Acceptance criteria should be:

- measurable,
- verifiable,
- value-oriented.

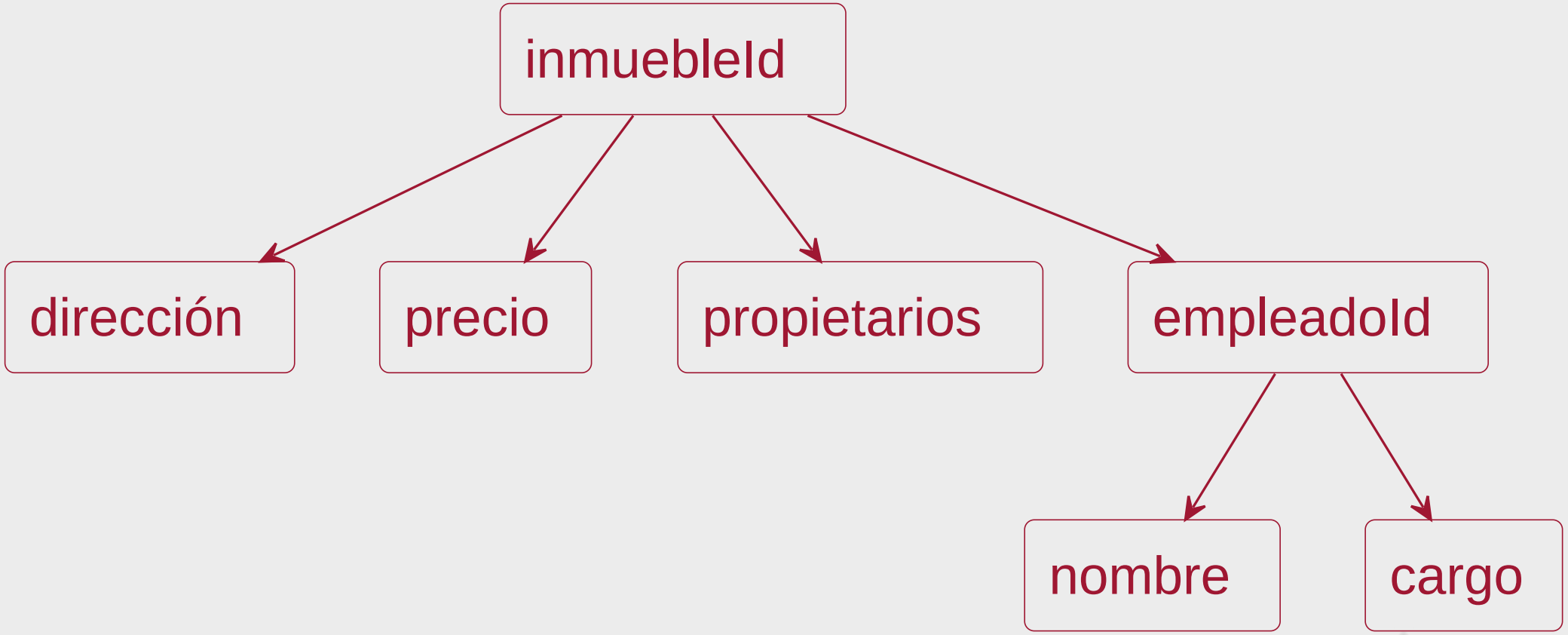


T6: UML conceptual model

Class, attribute, and association notation in UML.

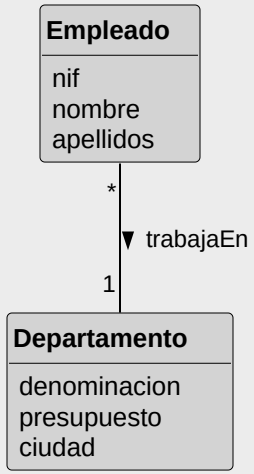
T7: functional dependencies

Dependency graph used to reason about candidate keys and normalization.



T8: conceptual-to-relational mapping

```
Departamento(id_dep, nombre)
Empleado(id_emp, nombre, FK id_dep)
```



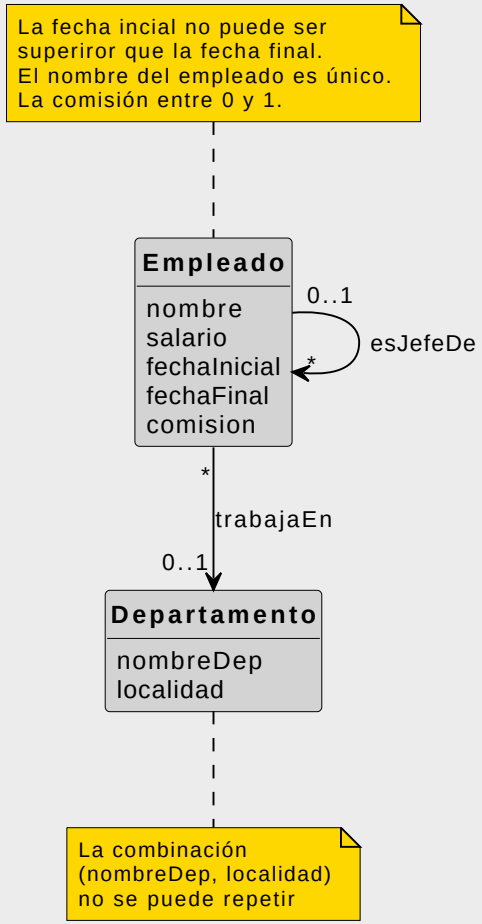
T9: relational algebra join

$$\Pi_{\text{nombre,salario}}\left(Empleado \bowtie_{\text{Empleado.id_dept=Departamento.id_dept}} Departamento\right)$$

$$\underset{R}{\begin{array}{cc} A & B \\ \hline a_1 & b_1 \\ a_2 & b_1 \\ a_3 & b_2 \end{array}} \bowtie \underset{S}{\begin{array}{cc} B & C \\ \hline b_1 & c_1 \\ b_2 & c_2 \\ b_3 & c_3 \end{array}} = \begin{array}{ccc} a_1 & b_1 & c_1 \\ a_2 & b_1 & c_1 \\ a_3 & b_2 & c_2 \end{array}$$

T10: SQL DDL and model

```sql
CREATE TABLE departamento (
  id_dept INT PRIMARY KEY,
  nombre  VARCHAR(40) NOT NULL
);
```



T11: stored procedures

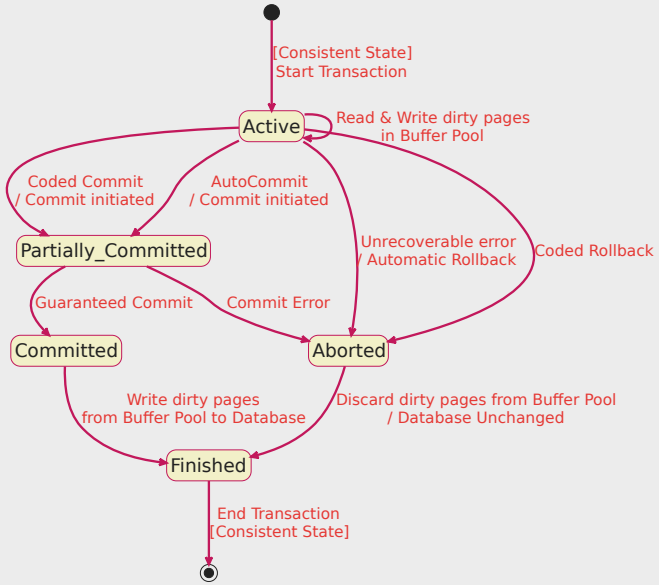
```
DELIMITER //
CREATE PROCEDURE p_raise_fee(IN p_pct DECIMAL(4,2))
BEGIN
  UPDATE empleado
  SET fee = fee * (1 + p_pct / 100);
END//
DELIMITER ;
```

```
CREATE
    [OR REPLACE]
    [DEFINER = { user | CURRENT_USER | role | CURRENT_ROLE }]
    PROCEDURE sp_name ([proc_parameter[,...]])
    [characteristic ...] routine_body

proc_parameter:
    [ IN | OUT | INOUT ] param_name type

type:
    Any valid MariaDB data type

characteristic:
    LANGUAGE SQL
  | [NOT] DETERMINISTIC
  | { CONTAINS SQL | NO SQL | READS SQL DATA | MODIFIES SQL DATA }
  | SQL SECURITY { DEFINER | INVOKER }
  | COMMENT 'string'

routine_body:
    Valid SQL procedure statement
```

T12: transaction states

Typical DBMS states:

- `Active`
- `Partially_Committed`
- `Committed`
- `Aborted`
- `Finished`



Guide map

Section 1. Typography and code fonts

Section 2. Base layouts and images

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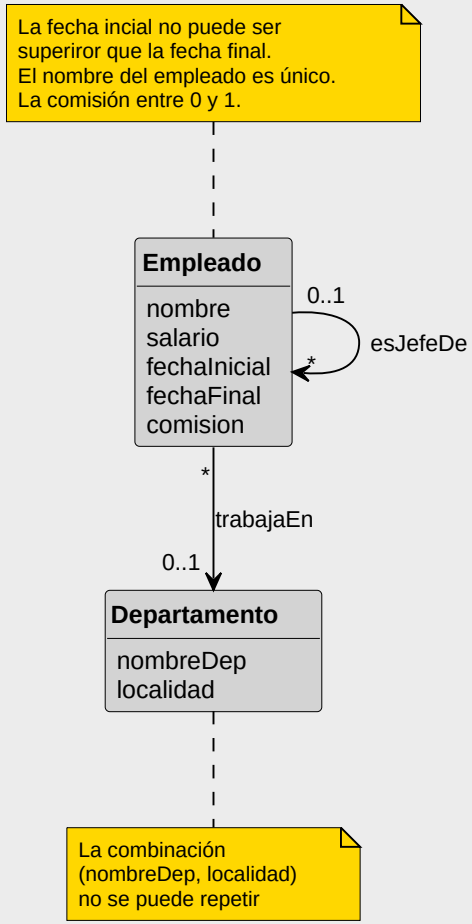
Section 4. `mr`, `sql`, and KaTeX

Section 5. Visual regression cases A-J

Section 6. Realistic topic examples T1-T12

Section 7. Mixed regression cases K-N

Test K: left image with right code




```
Departamentos = {departamentoId, nombreDep, localidad}
  PK(departamentoId)
  AK(nombreDep, localidad)

Empleados = {empleadoId, departamentoId, jefeId,
  nombre, salario, fechaInicial, fechaFinal, comision}
  PK(empleadoId)
  FK(departamentoId)/Departamentos
  FK(jefeId)/Empleados
```

Additional text below the code block.

Test L: left code with right image

Text above the code.

```
CREATE TABLE cuentas (
  numcta SMALLINT KEY,
  titular VARCHAR(20) NOT NULL,
  saldo DECIMAL(9,2) NOT NULL,
  CHECK (saldo >= 0)
);
```

Text below the code.

Test M: 50/50 image and code

Text above the image.

Text below the image.

Text above the code.

```
Pedido(id_ped, fecha, FK id_cli)
Cliente(id_cli, nombre, ciudad)
```

Text below the code.

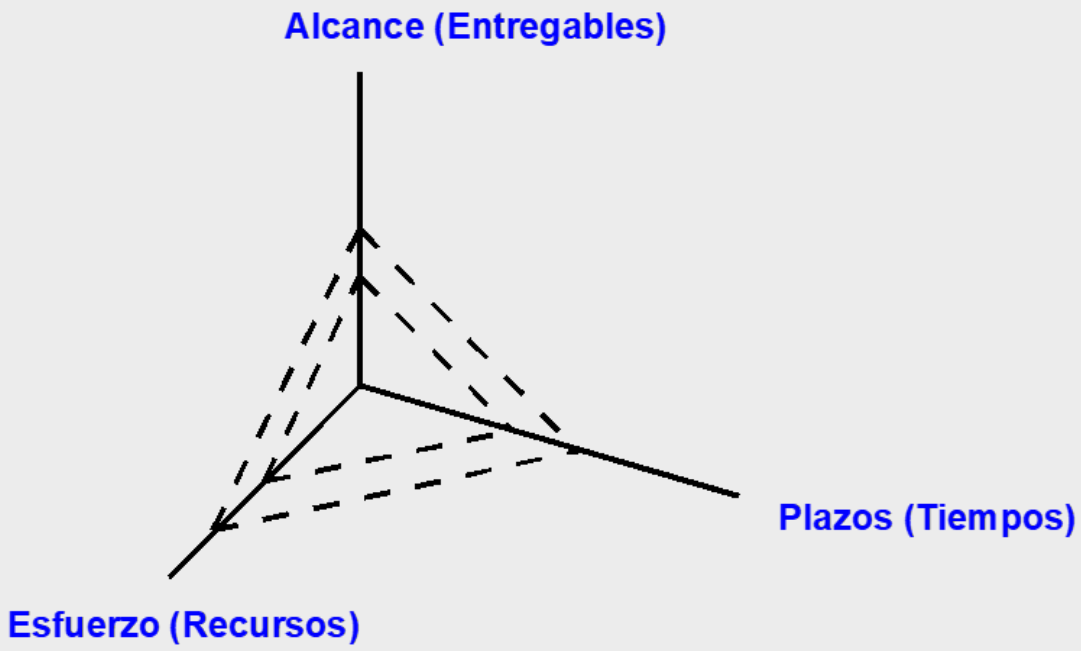
Test N: 50/50 code and image, inverted

Text above the code.

```
SELECT e.nombre, d.nombre
FROM Empleado e
JOIN Departamento d ON d.id_dept = e.id_dept;
```

Text below the code.

Text above the image.



Text below the image.